

Test Plan

Online Book Store System

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1. Introduction

This High-Level Test Plan defines the overall testing approach for the Online Book Store System, an e-commerce web application that enables users to register, browse books, search by title or author, view book details, manage a shopping cart, and purchase books online through a secure checkout process.

The purpose of this document is to establish the testing strategy, scope, objectives, environment, risks, deliverables, and execution approach that will be followed to verify that the application satisfies both its functional and selected non-functional requirements before release.

Testing will be performed primarily through **manual testing**, with test cases designed, managed, and executed using **Testworthy**. A risk-based testing approach will be followed to prioritize business-critical functionality such as authentication, shopping cart management, checkout, and payment processing. Stable and repetitive test cases will also be identified as candidates for future automation.

2. Test Strategy

Testing will follow a structured manual testing process based on the functional requirements defined during the Requirement Analysis phase.

The testing activities will be performed in the following sequence:

- Requirement Review
- Test Planning
- Test Case Design
- Test Suite Organization in Testworthy
- Test Execution
- Defect Reporting
- Regression Testing after defect fixes
- Test Execution Reporting
- QA Sign-off

Risk-based testing will be applied throughout the testing cycle, with the highest priority given to critical business workflows such as:

- User Registration and Authentication
- Shopping Cart Management
- Checkout and Payment Processing
- Order Confirmation

Smoke testing will be performed before detailed functional testing to verify build stability, while regression testing will be conducted after bug fixes to ensure existing functionality continues to operate correctly.

3. Scope of Testing

3.1 In Scope

The following features are included in this testing cycle.

Functional Testing

- User Registration
- User Login and Authentication

- Password Reset
- Browse Book Catalog
- Search Books
- View Book Details
- Shopping Cart Management
- Checkout Process
- Order Confirmation
- Order History

Non-Functional Validation

The following quality attributes will receive basic validation during this testing cycle:

- Performance sanity checks
- Browser compatibility
- Usability
- Input validation
- Basic security verification

3.2 Out of Scope

The following items are outside the scope of this testing cycle:

- Backend administration modules
- Inventory management system
- Internal payment gateway implementation
- Load, stress, and endurance testing
- Localization and multilingual support
- Native mobile applications
- Third-party integrations other than payment gateway validation

4. Testing Objectives

- Verify that all core functional requirements of the Online Book Store operate as specified.
- Identify and report defects in functional workflows, UI behavior, and input validation before release.
- Validate that critical business flows — registration, login, search, cart, and checkout — work end-to-end without data loss or incorrect calculations.
- Confirm that appropriate error handling and validation messages are shown for invalid or unexpected user input.
- Establish a reusable, traceable set of test cases in Testworthy that can support future regression and automation efforts.
- Provide management with clear, evidence-based test execution reports to support release decisions.

5. Test Environment

The following test environment will be used to execute and validate the Online Book Store System. It defines the hardware, software, tools, and test data required to perform testing consistently throughout the testing cycle.

Component	Details
Application Under Test	Online Book Store Web Application
Test Environment	QA/Staging Environment
Database	Dedicated QA Test Database
Browsers	Google Chrome (Latest), Mozilla Firefox (Latest), Microsoft Edge (Latest)
Operating Systems	Windows 10, Windows 11, macOS
Screen Resolution	1920 × 1080
Test Data	Pre-configured user accounts, book catalog, payment sandbox accounts
Test Management Tool	Testworthy
Defect Tracking	Organizational Bug Tracking Process (Assignment 3)
Internet Connection	Stable broadband connection

6. Testing Types

Different testing types will be performed to verify the quality, reliability, and stability of the Online Book Store System. Each testing type serves a specific purpose during the testing lifecycle.

Testing Type	Objective
Functional Testing	Verify that each feature behaves according to the documented requirements.
Smoke Testing	Validate that critical system functionality works correctly before detailed testing begins.
Regression Testing	Ensure previously tested functionality continues to work after changes or bug fixes.
Negative Testing	Verify that the application handles invalid, missing, or unexpected input correctly.
Usability Testing	Evaluate navigation, error messages, and overall ease of use.
Compatibility Testing	Verify application behavior across supported browsers and operating systems.
Basic Performance Sanity Testing	Confirm acceptable response times for commonly used pages under normal conditions.
Automation Candidate Review	Identify stable and repetitive test cases suitable for future automation.

7. Entry and Exit Criteria

7.1 Entry Criteria

- Functional requirements and user stories for the Online Book Store are finalized and reviewed.
- Test cases have been written, reviewed, and uploaded into Testworthy under appropriate test suites.
- The QA/staging test environment is available, stable, and deployed with the build to be tested.
- Required test data (user accounts, book catalog, payment sandbox) is available in the test environment.

7.2 Exit Criteria

- 100% of planned test cases have been executed (Pass, Fail, or Blocked) and results recorded in Testworthy.
- No open High-severity or blocking defects remain unresolved in core workflows (login, cart, checkout).
- At least 95% of Smoke test cases pass, and at least 90% of all planned test cases pass.
- All identified defects are logged with severity/priority and communicated to the development team.
- Test execution report and test case export have been generated and reviewed.

8. Risks and Assumptions

8.1 Risks

The following risks have been identified based on the scope of testing and project dependencies. These risks may affect the testing schedule, execution quality, or overall project delivery if not properly managed.

Risk	Business Impact	Mitigation
Test environment instability	Delays test execution and affect result reliability	Ensure QA environment is validated before execution
Third-party payment gateway downtime	Checkout testing may be blocked	Use sandbox or mocked payment responses where applicable
Requirement changes during testing	Existing test cases may require modification	Freeze requirements before execution whenever possible
Limited testing time	Lower-priority scenarios may not be executed	Prioritize testing based on business risk
Incomplete or incorrect test data	May reduce testing effectiveness	Prepare representative test data before execution

8.2 Assumptions

The following assumptions are made to ensure that testing can be executed effectively within the planned scope and schedule.

- A stable application build will be deployed before testing begins.
- Functional requirements have been reviewed and approved.
- Testworthy remains available throughout the testing cycle.
- Test accounts and payment sandbox credentials are available.
- Test data accurately represents production business scenarios.

9. Test Deliverables

The following assumptions are made to ensure that testing can be executed effectively within the planned scope and schedule.

Deliverable	Description
High-Level Test Plan	Overall testing strategy and approach
Test Cases	Test cases designed and organized in Testworthy
Test Suites	Organized collection of related test cases
Test Execution Results	Pass, Fail, and Blocked execution status

Test Execution Report	Summary report generated from Testworthy
CSV Export	Exported test cases with execution information
Defect Report	List of defects identified during testing
Final QA Summary	Testing completion summary and release recommendation

10. Non-Functional Requirements Considered

While the primary focus of this cycle is functional testing, the following quality attributes are noted for awareness and light validation where feasible within scope:

Quality Attribute	Validation Approach
Performance	Verify acceptable response time for catalog browsing and search pages.
Security	Validate input validation, authentication behavior, and secure handling of user data.
Availability	Confirm application remains accessible throughout testing.
Usability	Evaluate navigation, consistency, and clarity of error messages.
Compatibility	Verify application behavior across supported browsers.
Scalability	Documented for future performance testing; not fully evaluated during this cycle.

11. Test Execution Approach

Testing will be conducted in a structured sequence, beginning with requirement verification and test case preparation, followed by test execution in Testworthy. Defects identified during execution will be logged, tracked, and retested after resolution. Once all planned test cases have been executed and the exit criteria have been satisfied, a final test execution report will be generated and the testing cycle will be considered complete.

12. Conclusion

This High-Level Test Plan establishes the overall testing strategy for the Online Book Store System by defining the testing scope, objectives, environments, execution approach, risks, assumptions, and deliverables. It provides a structured framework for designing and executing manual test cases while ensuring traceability between requirements and testing activities. Following this plan will support consistent testing, effective defect identification, and informed release decisions based on measurable quality criteria.